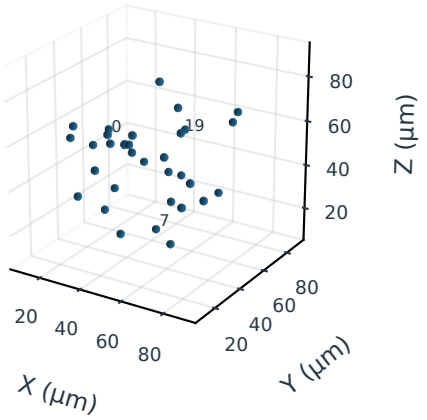
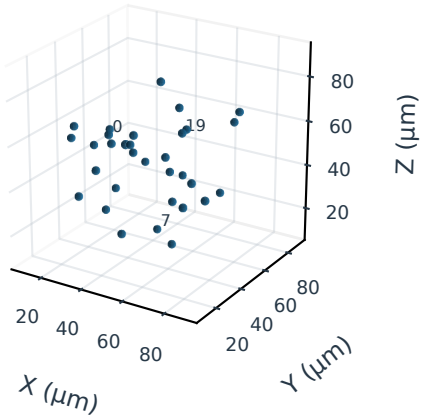
No-flow full-particle dynamics: OFF versus three PZT ON

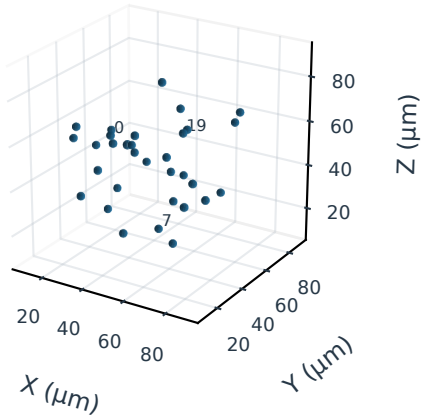
a Initial 32 bubbles



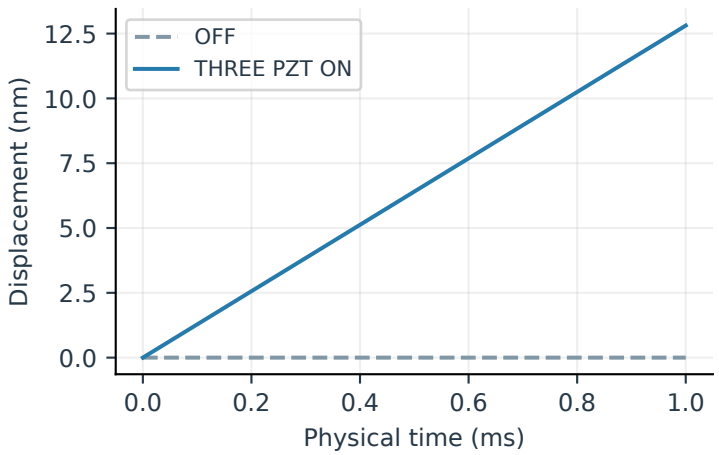
b OFF at 1 ms



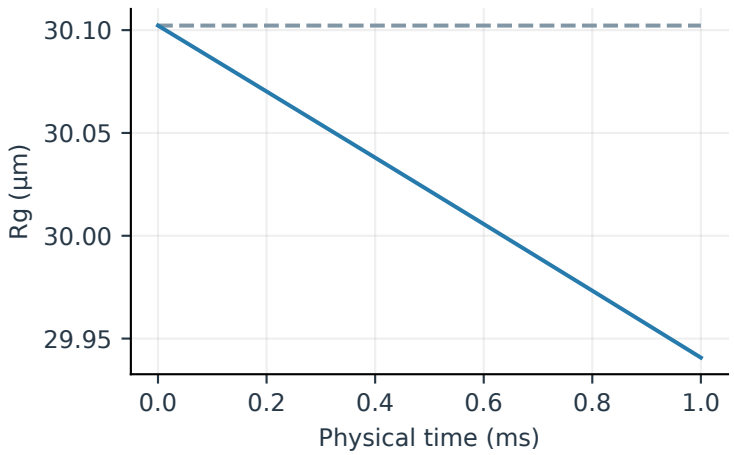
c Three PZT ON at 1 ms



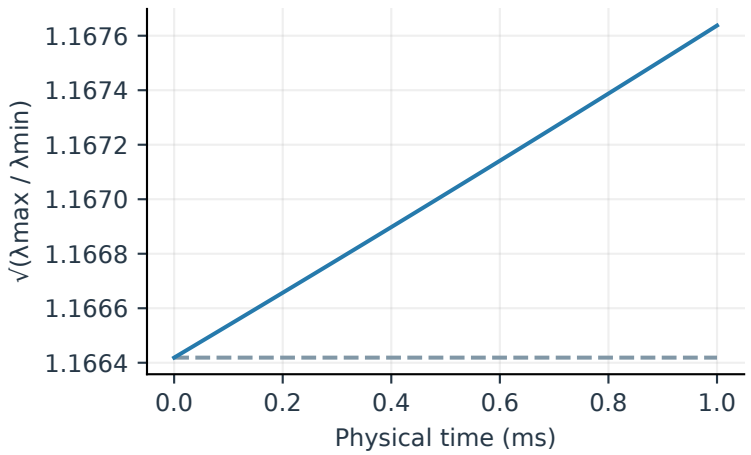
d Centroid displacement



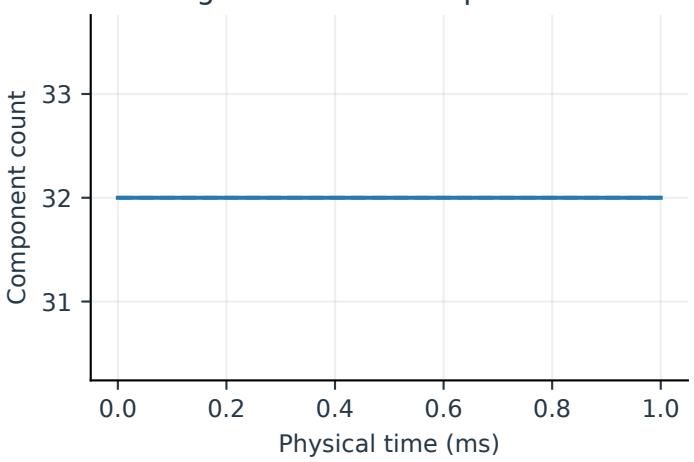
e Radius of gyration



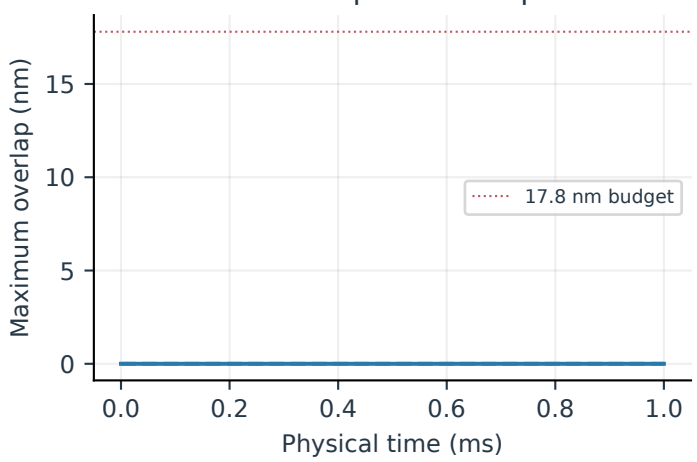
f Global aspect ratio



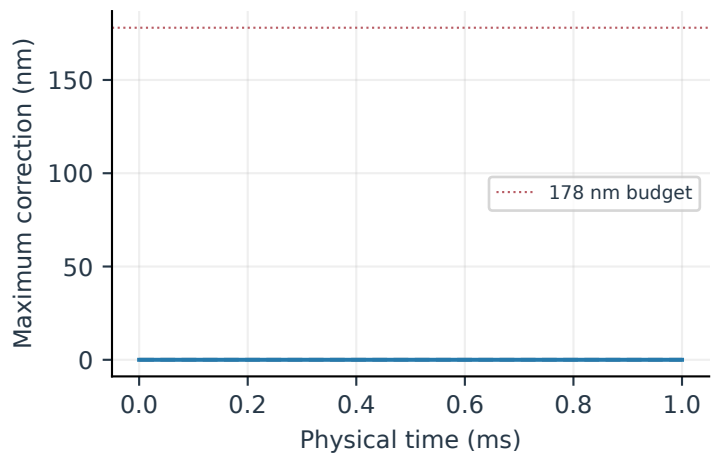
g Connected components



h Accepted overlap



i Wall correction



Fixed IDs; actual R = 1.78 μm and equal XYZ scales. Raw saved frames; no position or time smoothing.
ON local |P| peak: 1111.8–1412.3 Pa. Both trajectories: 1 ms, 1 μs steps. This short test does not establish aggregation or target control.